

Coding & Solution

Date	18 July 2026
Team ID	TBD
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/mani006281/Voice-Based-Concept-Understanding-Analyser
Programming Language(s)	Python
Framework(s) Used	Streamlit, SQLAlchemy

Key Features Implemented	• Voice Recording & Audio Upload • Speech-to-Text using Whisper • Semantic Similarity Analysis • AI Feedback using Gemini • Performance Dashboard • SQLite Database • PDF Report Generation
Pending / Incomplete Features	• User Authentication • Cloud Deployment
Setup / Run Instructions	1. Install Python 3.11+ 2. Install requirements using: pip install -r requirements.txt 3. Run: streamlit run frontend/streamlit_app.py



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

